CS728: Assignment 1

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Contents

Chapter 1

Task 1

The following table shows the final parameters used for the Glove training:

Hyperparameter	Value
Window Size (w)	10
Learning Rate (η)	0.05
Epochs	25
$x_{\max}$	100
α	0.75

Table 1.1: Selected GloVe hyperparameters

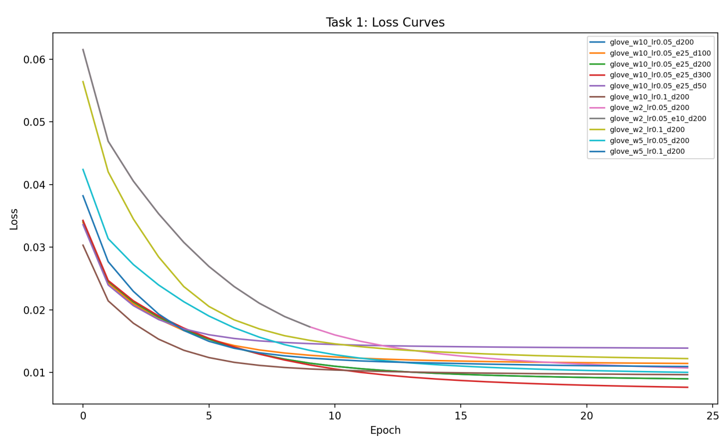

Figure 1.1: Loss curves for different embedding dimensions

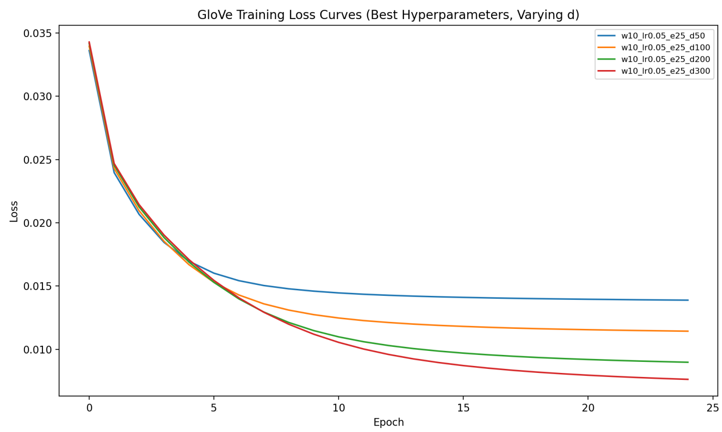

Figure 1.2: Loss curves for best hyperparameters across embedding dimensions

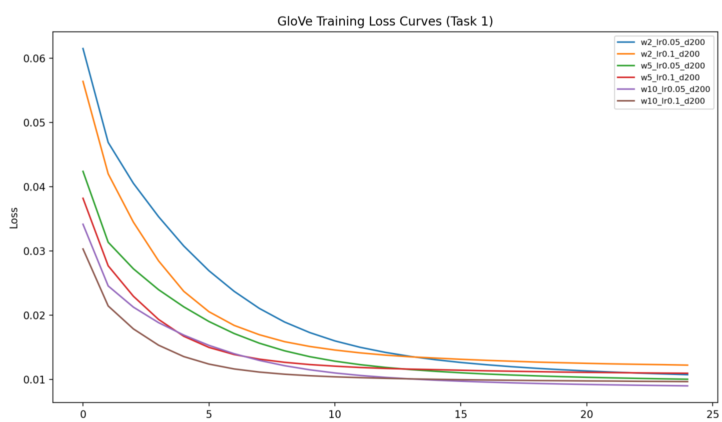

Figure 1.3: Loss curves for embedding dimension 200

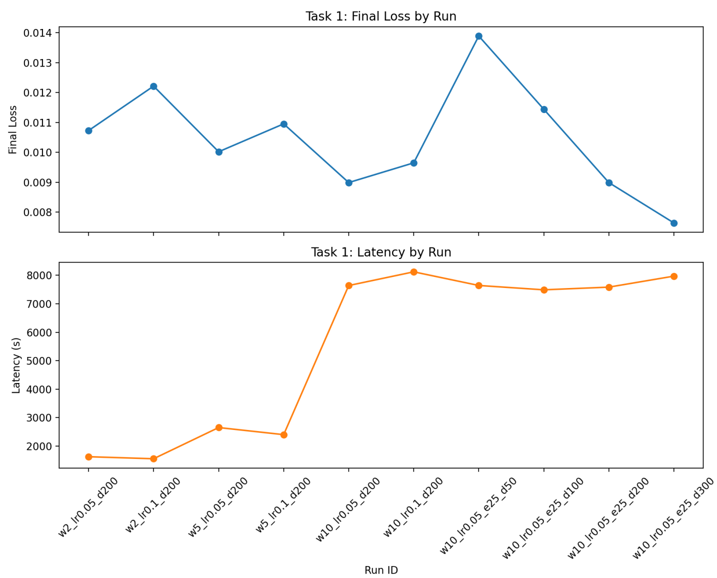

Figure 1.4: Final loss and Latency for different hyperparameters

Chapter 2

Task 2

Chapter 3

Task 3

Chapter 4

Task 4

We used a simple MLP architecture with a single hidden layer with 128 neurons followed by a ReLU and a dropout layer with a dropout rate of 0.2. The final output layer has 9 neurons corresponding to the 9 classes in the dataset. For OOV words we use the mean of the embeddings of the words in the training set as the embedding for the OOV word as it acts like a neutral embedding and does not bias the model towards any particular class. We trained the model for 10 epochs with a batch size of 1024 and a learning rate of 0.001 using the Adam optimizer. The tables below show the results of our model on the test set.

Dimension	Accuracy	Weighted F1	Macro F1
50	0.8309	0.7619	0.2179
100	0.8314	0.7643	0.2198
200	0.8314	0.7653	0.2224
300	0.8319	0.7658	0.2241

Table 4.1: Test performance of GloVe-MLP across embedding dimensions

Dimension	Accuracy	Weighted F1	Macro F1
50	0.8274	0.7523	0.2043
100	0.8301	0.7589	0.2091
200	0.8309	0.7609	0.2126
300	0.8305	0.7607	0.2118

Table 4.2: Test performance of SVD-MLP across embedding dimensions

From this we see that the best performance is achieved with the GloVe embeddings with a dimension of 300 and the SVD embeddings with a dimension of 200. **TODO Add reason why CRF performs better**

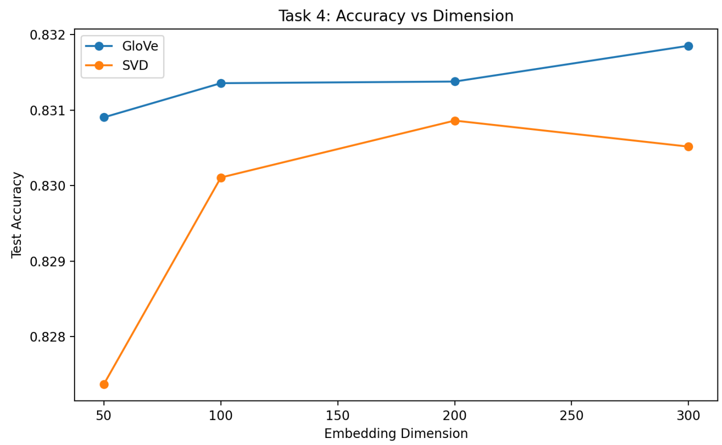

Figure 4.1: Test accuracy across embedding dimensions

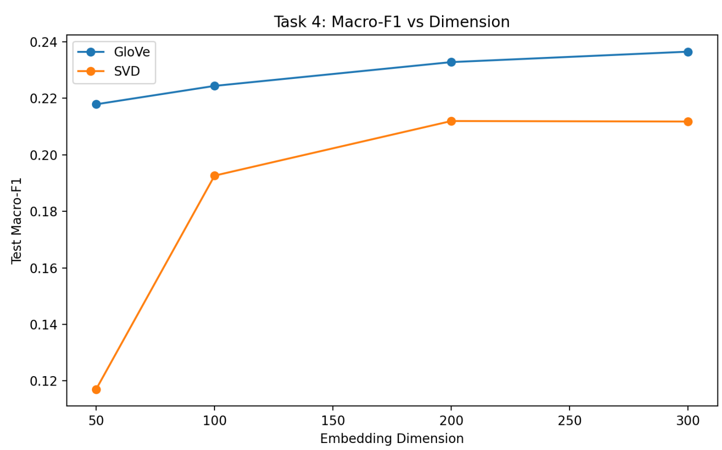

Figure 4.2: Test F1 score across embedding dimensions

Chapter 5

Task 5